

Theorem 1 (Pick's Theorem). *Let P be a simple polygon in the plane whose vertices all have integer coordinates. Let i be the number of lattice points in the interior of P , and let b be the number of lattice points on the boundary of P (including the vertices). Then the area A of P satisfies*

$$A = i + \frac{b}{2} - 1.$$

Proof. Add noncrossing diagonals until P is triangulated, using every lattice point of P as a vertex of the triangulation. Then each triangle in the triangulation is a primitive lattice triangle, meaning that it has no lattice points other than its three vertices on its boundary or in its interior. By the standard lattice-triangle lemma, every primitive lattice triangle has area $\frac{1}{2}$.

Let T be the number of triangles in the triangulation. Since area is additive,

$$A = \frac{T}{2}.$$

Now view the triangulation as a planar graph. Let V , E , and F be the numbers of vertices, edges, and faces. Because every lattice point of P is a vertex, we have

$$V = i + b.$$

The faces are the T bounded triangles together with the unbounded exterior face, so

$$F = T + 1.$$

Euler's formula gives

$$V - E + F = 2.$$

To count edges, let E_{int} be the number of interior edges. Then

$$E = E_{\text{int}} + b,$$

because the polygon boundary contributes exactly b edges, one between each pair of consecutive boundary lattice points. Counting sides of triangles gives

$$3T = 2E_{\text{int}} + b = 2E - b.$$

Hence

$$E = \frac{3T + b}{2}.$$

Substituting this into Euler's formula yields

$$(i + b) - \frac{3T + b}{2} + (T + 1) = 2,$$

so

$$T = 2i + b - 2.$$

Therefore

$$A = \frac{T}{2} = \frac{2i + b - 2}{2} = i + \frac{b}{2} - 1.$$

This is Pick's theorem.

□